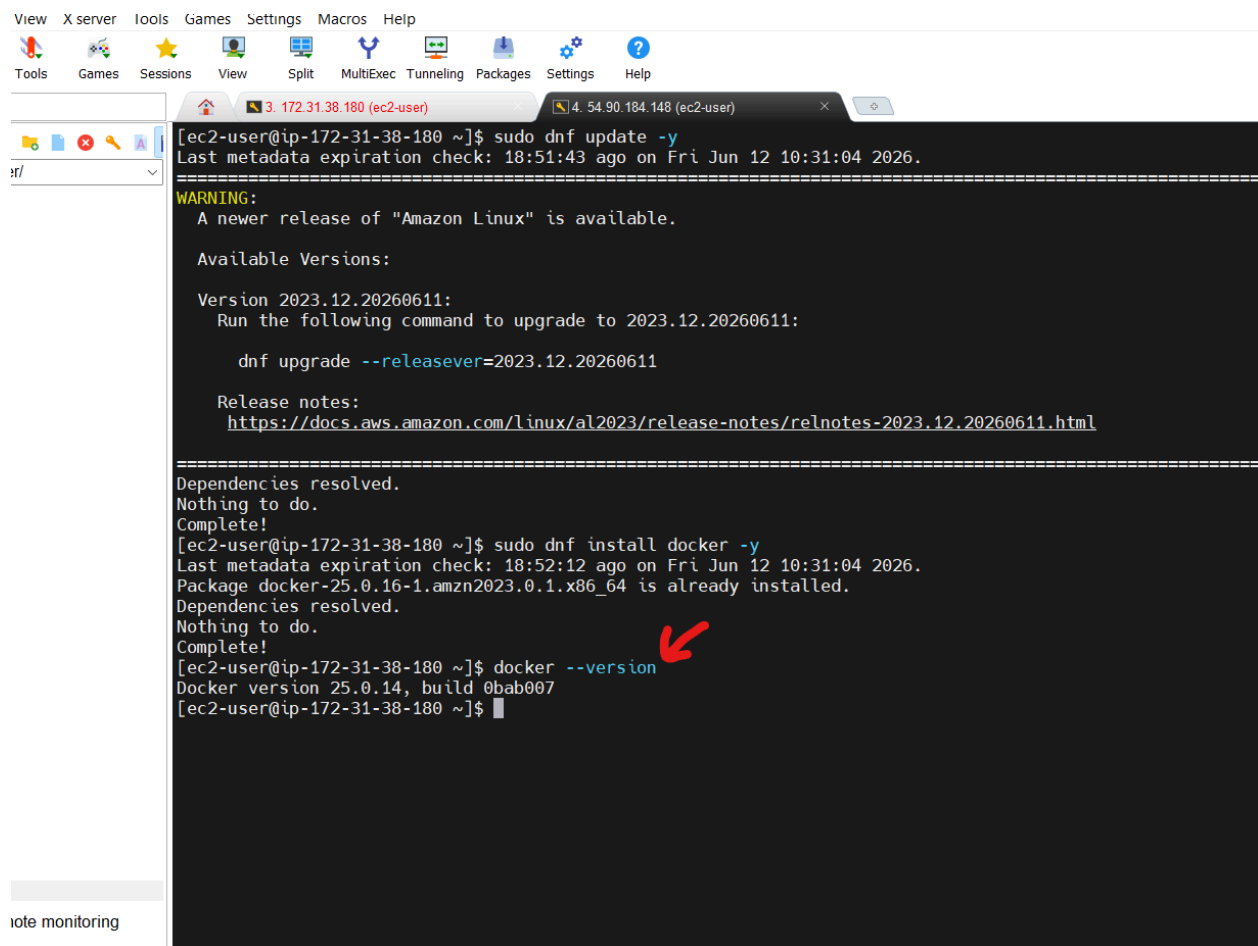


Name - Deepak Jha

Roll no - 24102B0055

CMPN B

Q1)



The screenshot shows a terminal window with a menu bar (View, X server, Tools, Games, Settings, Macros, Help) and a toolbar (Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, Help). The terminal output is as follows:

```
[ec2-user@ip-172-31-38-180 ~]$ sudo dnf update -y
Last metadata expiration check: 18:51:43 ago on Fri Jun 12 10:31:04 2026.

=====
WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.12.20260611:
Run the following command to upgrade to 2023.12.20260611:

dnf upgrade --releasever=2023.12.20260611

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20260611.html
=====

Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-38-180 ~]$ sudo dnf install docker -y
Last metadata expiration check: 18:52:12 ago on Fri Jun 12 10:31:04 2026.
Package docker-25.0.16-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-38-180 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-38-180 ~]$
```

A red arrow points to the `--version` flag in the `docker --version` command.

note monitoring

server Tools Games Settings Macros Help

Games Sessions View Split MultiExec Tunneling Packages Settings Help

3 172.31.38.180 (ec2-user) 4 54.90.184.148 (ec2-user)

```
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64
Verifying : nginx-1:1.30.2-1.amzn2023.0.1.x86_64
Verifying : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64
Verifying : nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch
Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

=====

WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.12.20260611:
Run the following command to upgrade to 2023.12.20260611:

dnf upgrade --releasever=2023.12.20260611

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20260611.html

=====

Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.30.2-1.amzn2023.0.1.x86_64                  nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64      nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[ec2-user@ip-172-31-38-180 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-38-180 ~]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   936fef290c8f   2 days ago    161MB
ubuntu       latest   30ba44506a6d   7 weeks ago    100MB
hello-world   latest   e2ac70e7319a   2 months ago  10.1kB
[ec2-user@ip-172-31-38-180 ~]$ sudo docker run -p
```

ip-172-31-38-180.ec2.internal 0% 0.39 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 19 hours ec2-user /: 28% /tmp: 0% /boot/efi: 13%

Please support Mahayarn by subscribing to the professional edition here: <https://mahayarn.mahatac.net>

Games Sessions View Split MultiExec Tunneling Packages Settings Help

3 172.31.38.180 (ec2-user) 4 54.90.184.148 (ec2-user)

```
Available Versions:

Version 2023.12.20260611:
Run the following command to upgrade to 2023.12.20260611:

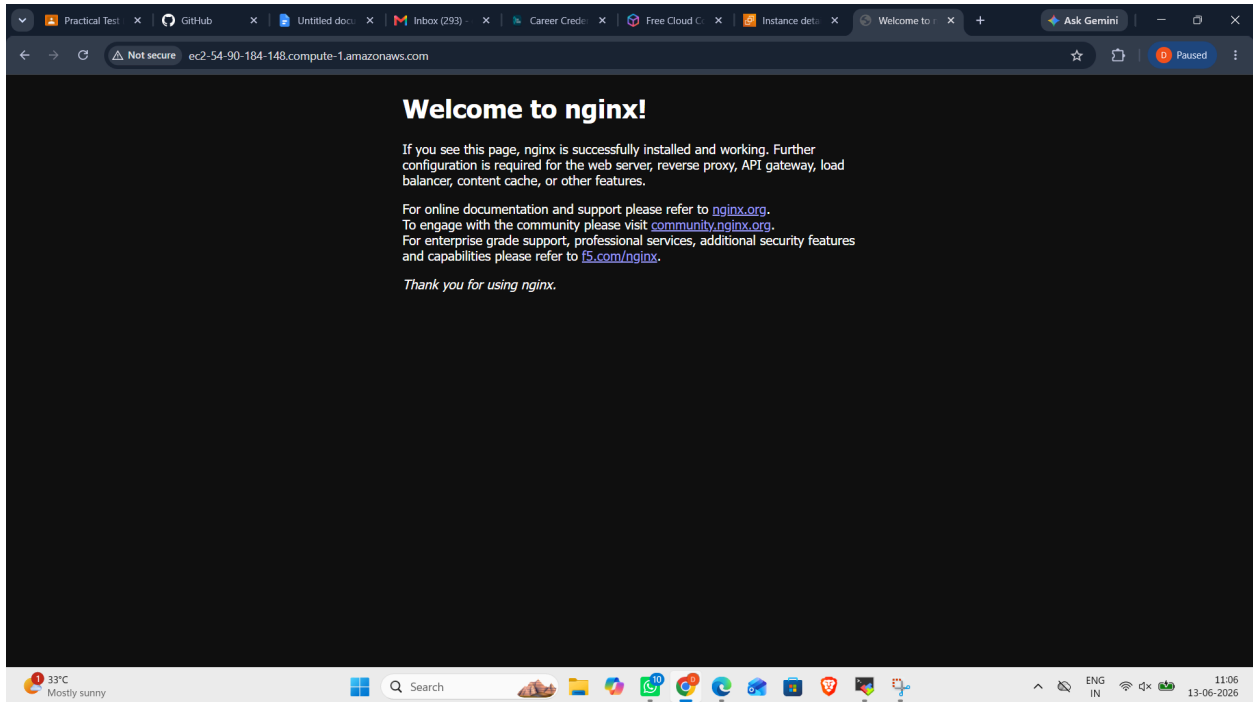
dnf upgrade --releasever=2023.12.20260611

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20260611.html

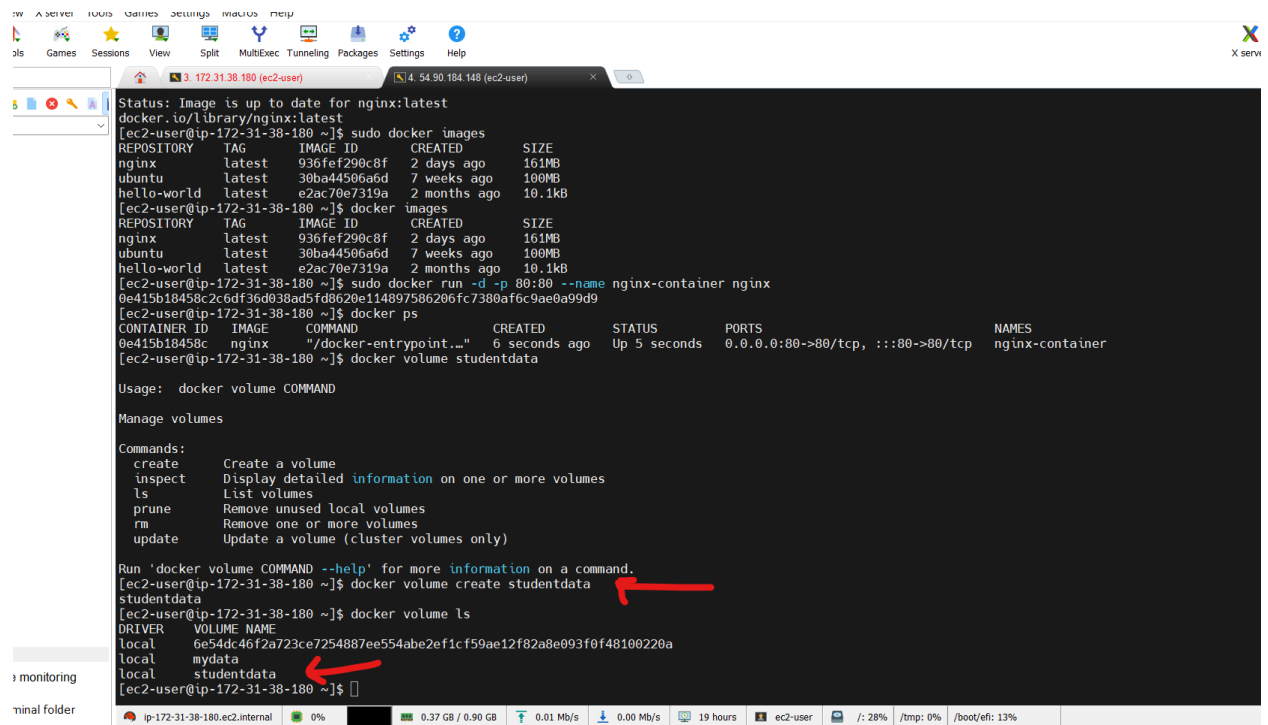
=====

Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.30.2-1.amzn2023.0.1.x86_64                  nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64      nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[ec2-user@ip-172-31-38-180 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-38-180 ~]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   936fef290c8f   2 days ago    161MB
ubuntu       latest   30ba44506a6d   7 weeks ago    100MB
hello-world   latest   e2ac70e7319a   2 months ago  10.1kB
[ec2-user@ip-172-31-38-180 ~]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   936fef290c8f   2 days ago    161MB
ubuntu       latest   30ba44506a6d   7 weeks ago    100MB
hello-world   latest   e2ac70e7319a   2 months ago  10.1kB
[ec2-user@ip-172-31-38-180 ~]$ sudo docker run -d -p 80:80 --name nginx-container nginx
0e415b18458c2c6df36d038ad5fd8620e114897586206fc7380af6c9ae8a99d9
[ec2-user@ip-172-31-38-180 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
0e415b18458c   nginx    "docker-entrypoint..." 6 seconds ago  Up 5 seconds  0.0.0.0:80->80/tcp, :::80->80/tcp  nginx-container
[ec2-user@ip-172-31-38-180 ~]$
```



Q2)



server Tools Games Settings Macros Help

Games Sessions View Split MultiExec Tunneling Packages Settings Help

3. 172.31.38.180 (ec2-user) 4. 54.90.184.148 (ec2-user)

```
[ec2-user@ip-172-31-38-180 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx          latest    936fef290c8f   2 days ago    161MB
ubuntu         latest    38ba44506a6d   7 weeks ago    100MB
hello-world    latest    e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-38-180 ~]$ sudo docker run -d -p 80:80 --name nginx-container nginx
0e415b18458c2c6df36d038ad5fd8620e114897586206fc7380af6c9ae0a99d9
[ec2-user@ip-172-31-38-180 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS      PORTS                               NAMES
0e415b18458c   nginx     "/docker-entrypoint...." 6 seconds ago  Up 5 seconds  0.0.0.0:80->80/tcp, :::80->80/tcp  nginx-container
[ec2-user@ip-172-31-38-180 ~]$ docker volume studentdata

Usage: docker volume COMMAND

Manage volumes

Commands:
  create      Create a volume
  inspect     Display detailed information on one or more volumes
  ls          List volumes
  prune       Remove unused local volumes
  rm          Remove one or more volumes
  update      Update a volume (cluster volumes only)

Run 'docker volume COMMAND --help' for more information on a command.
[ec2-user@ip-172-31-38-180 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-38-180 ~]$ docker volume ls
DRIVER    VOLUME NAME
local     6e54dc46f2a723ce7254887ee554abe2ef1cf59ae12f82a8e093f0f48100220a
local     mydata
local     studentdata
[ec2-user@ip-172-31-38-180 ~]$ sudo docker run -it --name ubuntu-container -v studentdata:/data ubuntu bash
root@d48129ba6597:/# touch /data/info.txt
root@d48129ba6597:/# echo "I am Deepak Jha" > /data/info.txt
root@d48129ba6597:/# echo "My Roll no is 24102B0055" >> /data/info.txt
root@d48129ba6597:/# cat /data/info.txt
I am Deepak Jha
My Roll no is 24102B0055
root@d48129ba6597:/#
```

Monitoring

Folder

ip-172-31-38-180.ec2.internal 0% 0.42 GB / 0.90 GB 0.01 Mb/s 0.00 Mb/s 19 hours ec2-user (x2) /: 28% /tmp: 0% /boot/efi: 13%

- Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

Q3)

The screenshot shows a MobaXterm session with two terminal windows. The top window displays the output of 'docker volume COMMAND' and 'docker volume ls'. The bottom window shows the output of 'docker buildx build' and 'docker images'.

```
Usage: docker volume COMMAND

Manage volumes

Commands:
  create   Create a volume
  inspect  Display detailed information on one or more volumes
  ls       List volumes
  prune    Remove unused local volumes
  rm       Remove one or more volumes
  update   Update a volume (cluster volumes only)

Run 'docker volume COMMAND --help' for more information on a command.
[ec2-user@ip-172-31-38-180 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-38-180 ~]$ docker volume ls
DRIVER      VOLUME NAME
local       6e54dc46f2a723ce7254887ee554abe2ef1cf59ae12f82a8e093f0f48100220a
local       mydata
local       studentdata
[ec2-user@ip-172-31-38-180 ~]$ sudo docker run -it --name ubuntu-container -v studentdata:/data ubuntu bash
root@d48129ba6597:/# touch /data/info.txt
root@d48129ba6597:/# echo "I am Deepak Jha" > /data/info.txt
root@d48129ba6597:/# echo "My Roll no is 24102B0055" >> /data/info.txt
root@d48129ba6597:/# cat /data/info.txt
I am Deepak Jha
My Roll no is 24102B0055
root@d48129ba6597:/# ^C
root@d48129ba6597:/# exit
exit
[ec2-user@ip-172-31-38-180 ~]$ mkdir practical-exam
[ec2-user@ip-172-31-38-180 ~]$ cd practical-exam
[ec2-user@ip-172-31-38-180 practical-exam]$ cat > Dockerfile << EOF
FROM ubuntu
CMD echo "Cloud Computing practical exam completed successfully"
EOF
[ec2-user@ip-172-31-38-180 practical-exam]$ cat Dockerfile
FROM ubuntu
CMD echo "Cloud Computing practical exam completed successfully"
[ec2-user@ip-172-31-38-180 practical-exam]$
```

```
root@d48129ba6597:/# ^C
root@d48129ba6597:/# exit
exit
[ec2-user@ip-172-31-38-180 ~]$ mkdir practical-exam
[ec2-user@ip-172-31-38-180 ~]$ cd practical-exam
[ec2-user@ip-172-31-38-180 practical-exam]$ cat > Dockerfile << EOF
FROM ubuntu
CMD echo "Cloud Computing practical exam completed successfully"
EOF
[ec2-user@ip-172-31-38-180 practical-exam]$ cat Dockerfile
FROM ubuntu
CMD echo "Cloud Computing practical exam completed successfully"
[ec2-user@ip-172-31-38-180 practical-exam]$ ^C
[ec2-user@ip-172-31-38-180 practical-exam]$ sudo docker build -t practical-image
ERROR: "docker buildx build" requires exactly 1 argument.
See 'docker buildx build --help'.

Usage: docker buildx build [OPTIONS] PATH | URL | -

Start a build
[ec2-user@ip-172-31-38-180 practical-exam]$ ^C
[ec2-user@ip-172-31-38-180 practical-exam]$ sudo docker build -t practical-image .
[*] Building 0.2s (5/5) FINISHED
==> [internal] load build definition from Dockerfile
==> transferring dockerfile: 174B
==> [internal] load metadata for docker.io/library/ubuntu:latest
==> [internal] load .dockerignore
==> transferring context: 2B
==> [1/1] FROM docker.io/library/ubuntu:latest
==> exporting to image
==> exporting layers
==> writing image sha256:4838a5dc4d9ef767f21540aadea3dd70f18dd39309df95e444b5c5ea0ad5e315
==> naming to docker.io/library/practical-image
[ec2-user@ip-172-31-38-180 practical-exam]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
practical-image latest    4838a5dc4d9e   7 weeks ago    100MB
ubuntu         latest    36ba4596a6d    7 weeks ago    100MB
hello-world    latest    e2ac70e7349a   2 months ago   10.1kB
[ec2-user@ip-172-31-38-180 practical-exam]$
```

viewA servertoolsGamesSettingsmanusHelp

ToolsGamesSessionsViewSplitMultiExecTunnelingPackagesSettingsHelp

3172.31.38.180 (ec2-user)4.54.90.184.148 (ec2-user)

X serverExit

er/practical-exam/

> EOF

[ec2-user@ip-172-31-38-180 practical-exam]\$ cat Dockerfile

FROM ubuntu

CMD echo "Cloud Computing practical exam completed successfully"

[ec2-user@ip-172-31-38-180 practical-exam]\$ ^C

[ec2-user@ip-172-31-38-180 practical-exam]\$ sudo docker build -t practical-image

ERROR: "docker buildx build" requires exactly 1 argument.

See 'docker buildx build --help'.

Usage: docker buildx build [OPTIONS] PATH | URL | -

Start a build

[ec2-user@ip-172-31-38-180 practical-exam]\$ ^C

[ec2-user@ip-172-31-38-180 practical-exam]\$ sudo docker build -t practical-image .

[+] Building 0.2s (5/5) FINISHED

[internal] load build definition from Dockerfile

=> transferring dockerfile: 174B

[internal] load metadata for docker.io/library/ubuntu:latest

[internal] load .dockerignore

=> transferring context: 2B

[1/1] FROM docker.io/library/ubuntu:latest

=> exporting to image

=> exporting layers

=> writing image sha256:4838a5dc4d9ef767f21540aadea3dd78f18dd39309df95e444b5c5ea8ad5e315

=> naming to docker.io/library/practical-image

[ec2-user@ip-172-31-38-180 practical-exam]\$ sudo docker images

REPOSITORYTAGIMAGE IDCREATEDSIZE

nginxlatest936fef290c8f2 days ago161MB

practical-imagelatest4838a5dc4d9e7 weeks ago100MB

ubuntulatest38ba44596a6d7 weeks ago100MB

hello-worldlateste2ac70e7319a2 months ago10.1kB

[ec2-user@ip-172-31-38-180 practical-exam]\$ ^C

[ec2-user@ip-172-31-38-180 practical-exam]\$ sudo docker run images

Unable to find image 'images:latest' locally

docker: Error response from daemon: pull access denied for images, repository does not exist or may require 'docker login': denied: requested access to the resource is denied.

See 'docker run --help'.

[ec2-user@ip-172-31-38-180 practical-exam]\$ sudo docker run practical-image

Cloud Computing practical exam completed successfully

[ec2-user@ip-172-31-38-180 practical-exam]\$

note monitoring

terminal folder

ip-172-31-38-180.ec2.internal0%0.40 GB / 0.90 GB0.01 Mb/s0.00 Mb/s19 hours

ec2-user/ : 28% /tmp: 0% /boot/efi: 13%